

Light-Cone Perturbation Theory and the Soft Gluon Wave Function

The CGC evolution operator Ω at next-to-leading order

[speaker]

Theory Seminar • September 15, 2026

based on **R. Radhakrishnan**, [arXiv:2607.18373](#) (JHEP submission)

Part I — Light-front dynamics

Quantisation on $x^+ = \text{const}$; why the vacuum is trivial; light-cone gauge; **LCPT** = old-fashioned perturbation theory done right.

Part II — Why small- x people care

Saturation in 10 minutes for a QCD expert: classical sources, Wilson lines, and why the **projectile wave function** is the object we need.

Part III — The paper

Construct the unitary boost operator Ω with $|\Psi_n\rangle = \Omega |n\rangle$ to $\mathcal{O}(g^2)$ in pure Yang–Mills:

- normal-ordered ansatz with **operator-valued** coefficients,
- fixed by **unitarity** + **matching to LCPT**,
- application: $\Omega^\dagger H \Omega$ is diagonal through $\mathcal{O}(g^2)$.

One-sentence summary

Ω is the operator that dresses a fast hadron with its soft gluon cloud — at leading order it is a coherent state, and **everything interesting starts at the next order**.

Part I — Light-front dynamics and LCPT

Dirac's forms of relativistic dynamics

Choose a foliation of spacetime; the generators of the Poincaré group split into **kinematical** (preserve the surface) and **dynamical** (evolve it).

Form	surface	# kinematical	Hamiltonian
Instant	$t = 0$	6	P^0
Front	$x^+ = (t + z)/\sqrt{2} = 0$	7	P^-
Point	$x^\mu x_\mu = a^2$	6	

The front form is the **maximally kinematical** one. In particular:

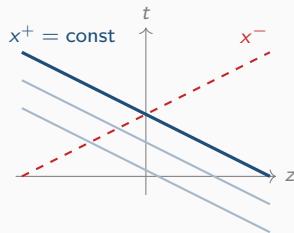
- **Boosts along z are kinematical:** $p^+ \rightarrow e^\omega p^+$, $\mathbf{p}$ untouched. A light-front wave function does *not* change shape under a longitudinal boost.
- Transverse boosts act as *Galilean* transformations in the $(\mathbf{p}, p^+)$ plane.

Why a high-energy physicist should care

Collider physics is about states boosted to enormous rapidity. In the front form the boost is trivial, and **parton distributions / LCWFs are frame-independent objects**.

$$x^{\pm} = \frac{1}{\sqrt{2}} (x^0 \pm x^3), \quad x^{\mu} = (x^+, x^-, \mathbf{x}), \quad p \cdot x = p^+ x^- + p^- x^+ - \mathbf{p} \cdot \mathbf{x}.$$

- x^+ = **light-cone time**; quantise on $x^+ = \text{const.}$
- p^- = **light-cone energy**, generates x^+ translations $\Rightarrow$ the Hamiltonian.
- p^+ = longitudinal momentum, **conserved and positive**.
- Measure $d^3 p \equiv dp^+ d^2 \mathbf{p}$,
 $\delta^3(k - p) \equiv \delta(k^+ - p^+) \delta^{(2)}(\mathbf{k} - \mathbf{p})$.



On shell ($m = 0$)

$$p^- = \frac{\mathbf{p}^2}{2p^+}$$

Non-relativistic in form: p^+ plays the role of a mass, $\mathbf{p}$ of a 2D momentum, p^- of the kinetic energy.

The one property that makes everything work

Positivity of $p^+ \Rightarrow$ trivial vacuum

For any physical quantum $p^+ > 0$, and p^+ is *conserved* at every vertex. A state with $P^+ = 0$ cannot fluctuate into anything with particles in it.

$$|0\rangle_{\text{free}} = |0\rangle_{\text{interacting}} \quad (\text{perturbatively})$$

Contrast with the equal-time formulation, where the vacuum is a complicated condensate of $q\bar{q}$ and gg pairs and the Fock expansion of a hadron is *not* meaningful.

Consequence: the Fock expansion is physical

$$|\Psi_h\rangle = \psi_0 |0\rangle + \psi_g |g\rangle + \psi_{gg} |gg\rangle + \psi_{ggg} |ggg\rangle + \dots$$

with $|\psi_n|^2$ a genuine **probability** to find that configuration in the hadron.

(The price: zero modes $p^+ \rightarrow 0$ hide the non-trivial QCD vacuum — chiral symmetry breaking, θ -vacuum. Not our problem today: we work with an explicit IR cutoff Λ on p^+ .)

- Only A^i , $i = 1, 2$ remain **dynamical**: two physical polarisations, no ghosts.
- A^- is a **constrained** field — solved from the $+$ component of the EOM,

$$\partial^+ A_+^a = -\frac{1}{\partial^+} \left(D_i^{ab} \partial^+ A_i^b - g J^{a+} \right),$$

and eliminated from the Hamiltonian.

- The price: **non-local $1/\partial^+$ kernels** $\rightarrow$ **instantaneous** interactions (light-front analogue of the Coulomb term in $A^0 = 0$ gauge).

Pure Yang–Mills LC Hamiltonian ($N_f = 0$)

$$H_{\text{YM}}^{\text{LC}} = \int dx^- d^2\mathbf{x} \left[\frac{1}{2} \Pi^a \Pi^a + \frac{1}{4} F_{ij}^a F_{ij}^a \right], \quad \Pi^a \equiv -\frac{1}{\partial^+} (D_i^{ab} \partial^+ A_i^b)$$

$$H_{\text{int}} = \int dx^- d^2\mathbf{x} \left[\underbrace{-gf^{abc} A_i^b A_j^c \partial_i A_j^a}_{3g \text{ vertex}} + \underbrace{\frac{g^2}{4} f^{abc} f^{ade} A_i^b A_j^c A_i^d A_j^e}_{4g \text{ vertex}} \right. \\ \left. \underbrace{-gf^{abc} (\partial_i A_i^a) \frac{1}{\partial^+} (A_j^b \partial^+ A_j^c) + \frac{g^2}{2} f^{abc} f^{ade} \frac{1}{\partial^+} (A_i^b \partial^+ A_i^c) \frac{1}{\partial^+} (A_j^d \partial^+ A_j^e)}_{\text{instantaneous}} \right]$$

Light-cone perturbation theory (LCPT)

LCPT = **old-fashioned (time-ordered) perturbation theory** on null planes. Split $H = H_0 + H_{\text{int}}$ and solve $H|\Psi\rangle = E|\Psi\rangle$ perturbatively:

$$|\Psi_n^{(1)}\rangle = \sum_{m \neq n} |m\rangle \frac{\langle m | H_{\text{int}} | n \rangle}{E_n - E_m}, \quad E_n = \sum_{i \in n} \frac{\mathbf{k}_i^2}{2k_i^+}.$$

Rules

1. Draw all **x^+ -orderings** of a graph.
2. $(p^+, \mathbf{p})$ conserved at each vertex; p^- is not.
3. Every intermediate state: all lines **on shell**, all $p^+ > 0$.
4. One **energy denominator** $(E_{\text{in}} - E_{\text{interm.}})^{-1}$ per intermediate state.
5. Add the **instantaneous** vertices.

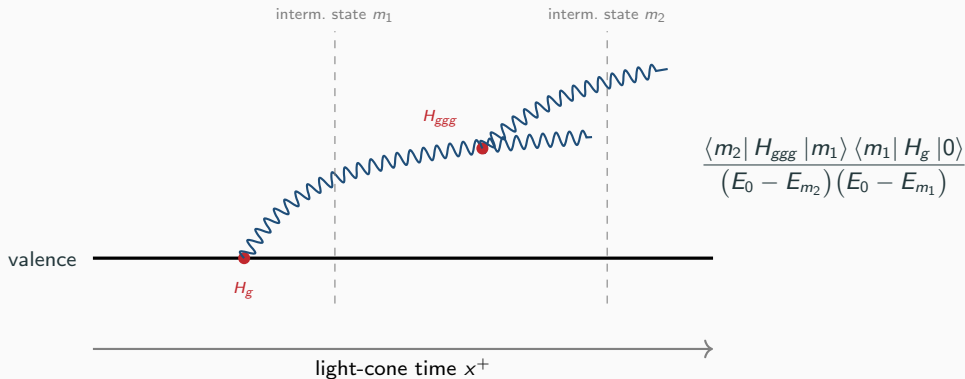
Relation to covariant diagrams

Do the k^- contour integral of a Feynman diagram by residues:

$$1 \text{ covariant graph} = \sum_{\substack{x^+ \text{-orderings} \\ \text{with all } p^+ > 0}} \text{LCPT graphs}$$

Orderings with a negative p^+ line collapse to the **instantaneous** pieces. Many orderings vanish $\Rightarrow$ fewer terms than naive OFPT.

Why use it here: each term is a *physical intermediate state* with a probabilistic reading — exactly what you need when the observable *is* a wave function.



- Dashed lines = where we “cut” and evaluate $E_{\text{in}} - E_{\text{interm.}}$.
- Each vertex is a matrix element of H_{int} between *on-shell* Fock states.
- Everything today is built out of exactly this object.

Part II — Small- x and the CGC in ten minutes

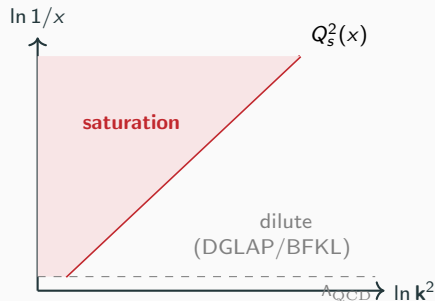
The small- x problem, stated for a QCD expert

- DGLAP/BFKL: the gluon density grows fast, $xG(x, Q^2) \sim x^{-\lambda}$, $\lambda \approx 0.3$.
- Occupation number of a mode $\mathbf{k}$ in a hadron of area $S_{\perp}$:

$$n(\mathbf{k}) \sim \frac{\alpha_s}{\mathbf{k}^2} \frac{xG(x, Q^2)}{S_{\perp}}$$

- Linear evolution $\Rightarrow n \sim 1/\alpha_s$: **fields are classically strong**, $A_{\mu} \sim 1/g$. Recombination $gg \rightarrow g$ can no longer be neglected.
- Balance defines the **saturation scale**

$$Q_s^2(x) \sim \alpha_s \frac{xG(x, Q_s^2)}{S_{\perp}} \gg \Lambda_{\text{QCD}}^2.$$



The effective theory

Colour Glass Condensate: separate the hadron by longitudinal momentum — **fast partons** = frozen colour sources ρ , **slow gluons** = the field they radiate.

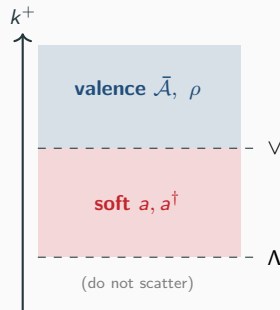
CGC: sources + field, and the Born–Oppenheimer logic

Split the gauge field by longitudinal momentum,

$$A_i^a = \underbrace{\bar{A}_i^a}_{k^+ \geq V \text{ (valence)}} + \underbrace{A_i^a}_{\Lambda < k^+ < V \text{ (soft)}}.$$

- Valence modes have much larger $k^+ \Rightarrow$ much **longer** LC time dilation: they are **static** on the soft time scale.
- Born–Oppenheimer: soft gluons see the valence sector only through its **colour charge density**

$$\rho^a(-\mathbf{p}) = -if^{abc} \int_V^\infty \frac{dk^+}{2\pi} \frac{d^2\mathbf{k}}{(2\pi)^2} a_j^{\dagger b}(k) a_j^c(k + \mathbf{p}).$$



The crucial non-Abelian twist

ρ is an **operator**, not a c-number: $[\rho^a(\mathbf{k}), \rho^b(\mathbf{p})] = if^{abc} \rho^c(\mathbf{k} + \mathbf{p})$.

Every coefficient below is a functional of $\rho \Rightarrow$ **ordering matters everywhere**.

The eikonal approximation and the soft Hamiltonian

Eikonal: drop everything suppressed by $k_{\text{soft}}^+/k_{\text{valence}}^+$. Operationally, $\frac{1}{\partial^+}$ acting on a *valence* field is dropped, $\frac{1}{\partial^+}(A_j^b \partial^+ \bar{A}_j^c) \simeq 0$.

After mode decomposition the interaction organises as

$$H_{\text{int}} = H_g + H_{ggg} + H_{gggg} + H_{gg\text{-inst}} + H_{gggg\text{-inst}} + H_V.$$

The key vertex: soft gluon emission off the source ($\mathcal{O}(g)$, one ρ)

$$H_g = \int_{\Lambda}^{\vee} \frac{dk^+}{2\pi} \frac{d^2\mathbf{k}}{(2\pi)^2} \frac{g k_i}{\sqrt{2} |k^+|^{3/2}} \left[a_i^{\dagger a}(k) \rho^a(-\mathbf{k}) + a_i^a(k) \rho^a(\mathbf{k}) \right]$$

- H_{ggg} — the genuine non-Abelian 3g vertex among *soft* gluons; ρ -independent.
- $H_{gg\text{-inst}}$ — valence instantaneously makes/absorbs a soft pair.
- H_V — valence-only; decouples from the soft dynamics.

Note the two **independent** labels: ($\#$ soft gluons n , $\#$ of ρ insertions m). A given order in g mixes different m .

What physics actually needs the wave function

Single inclusive gluon production at mid-rapidity

$$\frac{dN}{d^2\mathbf{k} dy} \propto \langle \Psi_{\text{proj}} | \hat{S}^\dagger a_i^\dagger(k) a_i(k) \hat{S} | \Psi_{\text{proj}} \rangle$$

- $|\Psi_{\text{proj}}\rangle = \Omega |v\rangle$: the projectile **boosted** by Ω .
- $\hat{S}$: eikonal scattering off the target field $\Rightarrow$ **Wilson lines** $a \rightarrow V^{ab} a^b$, $V = \mathcal{P} \exp[ig \int dx^- A_{\text{tgt}}^-]$.
- Average over the valence configurations ρ with the CGC weight.

So the calculation factorises into three ingredients, and **only one of them is hard**:

ingredient	status at LO	status at NLO
target Wilson lines	known	known (BK/JIMWLK)
eikonal $\hat{S}$	known	known
projectile wave function Ω	coherent state	this talk

Same Ω also generates the **JIMWLK** Hamiltonian when the soft modes are integrated out — its $\mathcal{O}(g^2)$ pieces feed NLO JIMWLK [Lublinsky–Mulian '17].

Part III — Constructing Ω at $\mathcal{O}(g^2)$

The evolution operator Ω : two equivalent definitions

(1) Dressing operator

$$|\Psi_n\rangle = \Omega |n\rangle, \quad \Omega = U(-\infty, 0)$$

Adiabatically switch on H_{int} : the free Fock state $|n\rangle$ evolves into the corresponding eigenstate of the full H . Physically, Ω is the boost by one rapidity slice: it fills the newly opened longitudinal phase space with soft gluons.

(2) Diagonalising operator

$$\Omega^\dagger H \Omega = \sum_n E_n |n\rangle \langle n|$$

Immediate from (1): $H\Omega |n\rangle = E_n\Omega |n\rangle$. So Ω is chosen to kill the off-diagonal matrix elements of H between Fock sectors, order by order.

Leading order: the coherent operator

Only H_g survives $\Rightarrow$

$$\Omega \rightarrow C = \exp \int \frac{d^3 p}{(2\pi)^3} \left[\mathcal{A}_j^b(p) a_j^{\dagger b}(p) - \mathcal{A}_j^{\dagger b}(p) a_j^b(p) \right], \quad \mathcal{A}_j^b(p) = -\frac{\sqrt{2} g p_j \rho^b(-\mathbf{p})}{\sqrt{p^+} \mathbf{p}^2}$$

i.e. the Weizsäcker–Williams field of the valence charge. $C|0\rangle$ is a coherent state: independently emitted WW gluons.

At fixed ρ , a coherent state has no connected correlations

$$\langle \hat{n}(\mathbf{k}_1) \hat{n}(\mathbf{k}_2) \rangle_\rho = \langle \hat{n}(\mathbf{k}_1) \rangle_\rho \langle \hat{n}(\mathbf{k}_2) \rangle_\rho$$

Every multi-gluon observable **factorises**. All LO correlations (double-inclusive, the near-side ridge) come *only* from the subsequent **averaging over ρ** .

Beyond LO, the normal-ordered expansion of Ω develops coefficients that are **genuinely connected multi-gluon emission amplitudes, present already before averaging over ρ** . They contribute to

- multiplicity **fluctuations** (moments of $\hat{N}$ at $\mathcal{O}(g^4)$),
- double-inclusive azimuthal correlations and the near-side ridge,
- NLO single-inclusive gluon production.

Goal of the paper

Determine **all** coefficients of the normal-ordered Ω through $\mathcal{O}(g^2)$ in pure Yang–Mills, completing partial results of Lublinsky–Mulian '17.

Step 1: the most general normal-ordered ansatz

Write Ω fully normal ordered in the soft $a, a^\dagger$, keeping every structure that can contribute through $\mathcal{O}(g^2)$:

$$\begin{aligned}\Omega = & N + \int_p \left[\mathcal{A}_1 a + \mathcal{A}_2 a^\dagger \right] + \int_{p,q} \left[B_1 aa + B_2 a^\dagger a^\dagger + B_3 a^\dagger a \right] \\ & + \int_{p,q,r} \left[C_1 a^\dagger a^\dagger a + C_2 a^\dagger aa \right] + \int_{p,q,r,s} \left[D_1 a^\dagger a^\dagger a^\dagger a + D_2 a^\dagger a^\dagger aa + D_3 a^\dagger aaa \right] \\ & + \int_{p,\dots,v} \left[E_1 a^\dagger a^\dagger a^\dagger a^\dagger aa + E_2 a^\dagger a^\dagger a^\dagger aaa + E_3 a^\dagger a^\dagger aaaa \right]\end{aligned}$$

(colour, polarisation and momentum labels suppressed; $\int_p \equiv \int d^3p/(2\pi)^3$.)

- $\mathcal{A}, C$ are $\mathcal{O}(g)$;
- $N - 1, B, D, E$ are $\mathcal{O}(g^2)$;
- all coefficients are **operators** built from ρ — they do not commute.

Reading the structures

$a^\dagger$: real emission. a : absorption.

$a^\dagger a$: propagation through the background.

$a^\dagger a^\dagger a$: $1 \rightarrow 2$ splitting of an existing gluon.

Step 2: two complementary sets of conditions

(a) Unitarity $\Omega^\dagger \Omega = 1$

Write $\Omega = 1 + g\Omega^{(1)} + g^2\Omega^{(2)} + \dots$

$$\mathcal{O}(g) : \quad \Omega^{(1)\dagger} = -\Omega^{(1)}$$

$$\mathcal{O}(g^2) : \quad \Omega^{(2)} + \Omega^{(2)\dagger} = -\Omega^{(1)\dagger} \Omega^{(1)}$$

Fixes precisely the pieces that matching *cannot* see:
the vacuum normalisation N , and the “ a -heavy”
structures $\mathcal{A}_1, C_2, B_1, B_3, D_2, D_3, E_2, E_3$.

(b) Matching to LCPT

Demand $\Omega |n\rangle = |\Psi_n\rangle$ sector by sector in Fock space, with $|\Psi_n\rangle$ computed **directly from H_{YM}^{LC}** .

A structure with m annihilation operators is probed *only* by an incoming state with $\geq m$ gluons $\Rightarrow$ must climb the Fock ladder:

incoming	fixes
$ 0\rangle$	$N, \mathcal{A}, B_2$
$ g\rangle$	B_3, C, D_1
$ gg\rangle$	D_2, E_1
$ ggg\rangle$	E_2

Unitarity + matching = all coefficients through $\mathcal{O}(g^2)$.

Order g (anti-Hermiticity of $\Omega^{(1)}$):

$$\mathcal{A}_{1j}^{\dagger b}(p) = -\mathcal{A}_{2j}^b(p) \equiv -\mathcal{A}_j^b(p), \quad C_2^{dc b}(p, q, r) = -C_1^{b c d}(r, q, p).$$

Order g^2 : e.g. the vacuum normalisation

$$N^2 = 1 - \int \frac{d^3 p}{(2\pi)^3} \mathcal{A}_j^{\dagger b}(p) \mathcal{A}_j^b(p) \implies \boxed{N = 1 - \frac{1}{2} \int \frac{d^3 p}{(2\pi)^3} \mathcal{A}^\dagger \mathcal{A}}$$

and, schematically,

$$B_1 = \frac{1}{2} \left[-2B_2^\dagger + \{\mathcal{A}^\dagger, \mathcal{A}^\dagger\} + 2 \int_r \mathcal{A}^\dagger C^\dagger \right], \quad B_3 + B_3^\dagger = -\{\mathcal{A}, \mathcal{A}^\dagger\} - \int_r (\mathcal{A}^\dagger C + C^\dagger \mathcal{A}) - \int_{r,s} C^\dagger C$$
$$D_2^\dagger + D_2 = -\mathcal{A} C^\dagger - \mathcal{A}^\dagger C - C^\dagger \mathcal{A} - C \mathcal{A}^\dagger - \int C^\dagger C, \quad E_3 = -E_1^\dagger + C^\dagger C^\dagger, \quad \dots$$

Physics content

Probability conservation: the virtual (a -heavy, number-preserving) coefficients are the “ $-\frac{1}{2}|\text{real}|^2$ ” partners of the real emission amplitudes. Note the **anticommutators** — a direct consequence of $[\rho, \rho] \neq 0$.

Step 3: act with Ω on Fock states

Vacuum

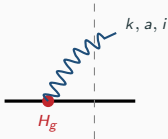
$$\Omega |0\rangle = N |0\rangle + (2\pi)^{3/2} \int_p \mathcal{A}_j^b(p) |g_j^b(p)\rangle + (2\pi)^3 \int_{p,q} B_{2kj}^{cb}(p,q) |g_j^b(p)g_k^c(q)\rangle$$

One gluon

$$\Omega |g_i^a(k)\rangle = \underbrace{-\frac{\mathcal{A}_i^{\dagger a}(k)}{(2\pi)^{3/2}} |0\rangle}_{\text{absorption}} + \underbrace{N |g_i^a(k)\rangle + \int_p B_{3ji}^{ba} |g_j^b(p)\rangle}_{\text{propagation in the background}} + \underbrace{(2\pi)^{3/2} \int_{p,q} C_{kji}^{cba} |g_k^c g_j^b\rangle}_{1 \rightarrow 2 \text{ splitting}} + \underbrace{(2\pi)^3 \int D_{1lkji}^{dcba} |ggg\rangle}_{1 \rightarrow 3}$$

- Normal ordering makes this mechanical: each a either **contracts with an incoming gluon** or kills the vacuum.
- Structures with more a 's than incoming gluons simply drop out $\Rightarrow$ the **triangular** structure of the matching table.
- Two- and three-gluon incoming states give long but equally mechanical expressions (Bose symmetrisation over the incoming legs) $\rightarrow$ fix D_2, E_1, E_2 .

Step 4: the LCPT side — $\mathcal{O}(g)$ and the WW field



$$|\Psi_{g\rho}^{LO}\rangle = - \int_{\Lambda}^{\vee} \frac{dk^+}{\sqrt{k^+}} \int d^2\mathbf{k} |g_i^a(k)\rangle \frac{g k_i}{2\pi^{3/2}\mathbf{k}^2} \rho^a(-\mathbf{k})$$

= the Weizsäcker–Williams field. Matching to $\Omega|0\rangle$:

$$\mathcal{A}_j^b(p) = - \frac{\sqrt{2} g p_j \rho^b(-\mathbf{p})}{\sqrt{p^+} \mathbf{p}^2}$$

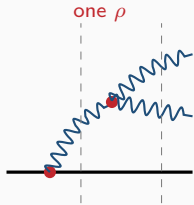
And the normalisation — where the small- x log lives

$$N = 1 - \frac{g^2}{8\pi^3} \int_{\Lambda}^{\vee} \frac{dp^+}{p^+} \int d^2\mathbf{p} \frac{\rho^a(-\mathbf{p})\rho^a(\mathbf{p})}{\mathbf{p}^2} = 1 - \frac{g^2}{8\pi^3} \ln \frac{\vee}{\Lambda} \int d^2\mathbf{p} \frac{\rho^a(-\mathbf{p})\rho^a(\mathbf{p})}{\mathbf{p}^2}$$

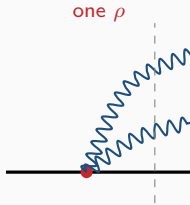
The $\ln(\vee/\Lambda) = \Delta y$ is precisely the **rapidity logarithm** resummed by BFKL/JIMWLK. Ω is the boost; iterating it generates the evolution equation.

Step 4 (cont.): $\mathcal{O}(g^2)$, two outgoing gluons

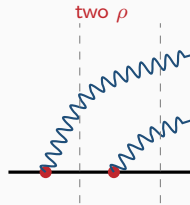
Three topologies, distinguished by how many ρ 's they carry:



(a) H_g then H_{ggg}



(b) $H_{gg\text{-}inst}$



(c) $H_g \times H_g$

The decomposition procedure

$$\begin{aligned}
 \text{(c) gives } \rho^b \rho^a &= \frac{1}{2} \underbrace{[\rho^b, \rho^a]}_{= \frac{i}{2} f^{bad} \rho^d(-\mathbf{k}) \Rightarrow \text{one } \rho} + \frac{1}{2} \underbrace{\{\rho^b, \rho^a\}}_{\text{two } \rho}
 \end{aligned}$$

The commutator part = two emitters at the *same* transverse point; colour algebra collapses it to a single effective source. The anticommutator is the genuine **two-source** piece — this makes the ρ -counting unambiguous.

Results I: the $\mathcal{O}(g)$ coefficients

Emission amplitude (one ρ)

$$\mathcal{A}_j^b(p) = -\frac{\sqrt{2} g p_j \rho^b(-\mathbf{p})}{\sqrt{p^+} \mathbf{p}^2}, \quad N = 1 - g^2 \int \frac{d^3 p}{(2\pi)^3} \frac{\rho^b(-\mathbf{p}) \rho^b(\mathbf{p})}{p^+ \mathbf{p}^2}$$

Splitting amplitude $a^\dagger a^\dagger a$ — pure $3g$ vertex, no ρ

$$C_{kji}^{cba}(k, p, q) = \frac{igf^{abc} (2\pi)^{9/2} \delta^{(3)}(k - p - q)}{8\pi^{3/2} \sqrt{k^+ p^+ q^+} \left(\frac{\mathbf{k}^2}{k^+} - \frac{\mathbf{p}^2}{p^+} - \frac{\mathbf{q}^2}{q^+} \right)} \left[\cdots \text{Lipatov-like vertex} \cdots \right]$$
$$[\cdots] = \left(p_i - q_i + \frac{q^+ - p^+}{k^+} k_i \right) \delta_{jk} + \left(k_j + q_j - \frac{k^+ + q^+}{p^+} p_j \right) \delta_{ik} + \left(\frac{k^+ + p^+}{q^+} q_k - p_k - k_k \right) \delta_{ij}$$

- $\mathcal{A}$ is the classical WW field — one power of ρ , $\propto 1/g$ when $\rho \sim 1/g$.
- C carries no ρ : it is the soft gluon splitting *in vacuum*, i.e. the seed of the non-linear, non-Abelian structure.
- Both are needed at $\mathcal{O}(g)$ — the coherent operator alone is incomplete.

Results II: the $\mathcal{O}(g^2)$ coefficients

All of $B_2, B_3, D_1, D_2, E_1, E_2$ are obtained in closed form. Structurally:

Two-gluon emission B_2 ($a^\dagger a^\dagger$)

two independent WW emissions + WW emission then $3g$ splitting + instantaneous pair creation off ρ .

Gluon-number preserving B_3 ($a^\dagger a$)

The **instantaneous** valence-gluon exchange with its Θ -function kinematic window, the $\{\rho, \rho\}$ Weizsäcker–Williams piece, and the H_{ggg} loop.

Higher coefficients: **irreducible vertex + products of lower coefficients**

$$D_1 = \underbrace{\frac{[4\text{-gluon vertex}]}{E_k - E_{pqr}}}_{\text{irreducible}} + \underbrace{\frac{C \mathcal{A}}{(\dots)} + \frac{\mathcal{A} C}{(\dots)} + \int_v \frac{C C}{(\dots)}}_{\text{reducible}}, \quad D_2 = \frac{[4\text{-gluon}]}{(\dots)} + \int_{r'} \frac{C C^\dagger}{(\dots)} + \frac{C \mathcal{A}^\dagger + \mathcal{A}^\dagger C}{(\dots)}$$

$$E_1 = \frac{C C}{(\dots)}, \quad E_2 = \frac{C C^\dagger}{(\dots)} \quad \implies \quad \text{the six-operator structures are **entirely reducible**.}$$

Resumming: the generator G

Exponentiate to expose the connected content:

$$\Omega = \exp(iG), \quad G = G^\dagger$$

$$\begin{aligned} G = & -i \int_p [\mathcal{A} a^\dagger - \mathcal{A}^\dagger a] - \frac{i}{2} \int_{p,q} [B_1 - B_2^\dagger] aa - \frac{i}{2} \int_{p,q} [B_2 - B_1^\dagger] a^\dagger a^\dagger - \frac{i}{2} \int_{p,q} [B_3 - B_3^\dagger] a^\dagger a \\ & - i \int_{p,q,r} [C a^\dagger a^\dagger a - C^\dagger a^\dagger aa] - \frac{i}{2} \int [D_1 - D_3^\dagger] a^\dagger a^\dagger a^\dagger a - \frac{i}{2} \int [D_2 - D_2^\dagger] a^\dagger a^\dagger aa + \dots \end{aligned}$$

- G is the **generator of the infinitesimal light-cone boost**; $\Omega = e^{iG}$ promotes it to a finite boost, resumming repeated emissions and virtual corrections.
- Hermiticity of G is automatic once the unitarity constraints are imposed — a strong **cross-check** on the whole calculation.
- $\mathcal{A}, C$ enter at $\mathcal{O}(g)$; B, D, E at $\mathcal{O}(g^2)$; B_1, D_3, E_3 are fixed purely by unitarity.

Application: diagonalising the soft Hamiltonian at $\mathcal{O}(g)$

Take $H_1 = H_0 + H_g + H_{ggg}$ and compute $\Omega^\dagger H_1 \Omega$:

$$\Omega^\dagger H_1 \Omega = H_0 + H_g + H_{ggg} + \int_p \left[-\mathcal{A} \frac{\mathbf{p}^2}{2p^+} a^\dagger + \mathcal{A}^\dagger \frac{\mathbf{p}^2}{2p^+} a \right] + \int_{p,q,r} \left[C \left(\frac{\mathbf{r}^2}{2r^+} + \frac{\mathbf{q}^2}{2q^+} - \frac{\mathbf{p}^2}{2p^+} \right) a^\dagger a^\dagger a - C^\dagger(\dots) a^\dagger a a \right]$$

Insert the explicit $H_g, H_{ggg}, \mathcal{A}, C$: every off-diagonal piece cancels identically.

$$\boxed{\Omega^\dagger H_1 \Omega = H_0}$$

The point

Earlier treatments diagonalised with the coherent operator C_{coh} alone (the WW field). That is **not sufficient**: the non-Abelian $3g$ vertex generates an off-diagonal structure at the *same* order, and one needs the $a^\dagger a^\dagger a$ operator C as well. With both, the diagonalisation at $\mathcal{O}(g)$ is **exact**.

Diagonalisation at $\mathcal{O}(g^2)$

Now keep everything: $H = H_0 + H_g + H_{ggg} + H_{gggg} + H_{gg\text{-inst}} + H_{gggg\text{-inst}}$.

$$\Omega^\dagger H \Omega = H_0 - \int_p \mathcal{A}_j^{\dagger b}(p) \mathcal{A}_j^b(p) \frac{\mathbf{p}^2}{2p^+} + 2 \int_{p,q,r,s} \left(\frac{\mathbf{r}^2}{2r^+} + \frac{\mathbf{s}^2}{2s^+} - \frac{\mathbf{p}^2}{2p^+} \right) C^\dagger C a^\dagger a$$

- The $C^\dagger C$ term vanishes identically — explicit evaluation gives 0. No residual off-diagonal (or $a^\dagger a$) contribution from the three-gluon sector.
- The coherent term survives:

$$\int \frac{d^3 p}{(2\pi)^3} \mathcal{A}^\dagger \mathcal{A} \frac{\mathbf{p}^2}{2p^+} = \int \frac{d^3 p}{(2\pi)^3} \frac{g^2 \rho^b(\mathbf{p}) \rho^b(-\mathbf{p})}{(p^+)^2}$$

Final result

$$\Omega^\dagger H \Omega = H_0 - \int \frac{d^3 p}{(2\pi)^3} \frac{g^2 \rho^b(\mathbf{p}) \rho^b(-\mathbf{p})}{(p^+)^2}$$

The only correction through $\mathcal{O}(g^2)$ is the **classical Weizsäcker–Williams background-field energy** $\propto \rho^2$ — exactly what one expects for a hadron dressed by its own coherent gluon cloud.

What was actually new here

Technical

- All coefficients of the normal-ordered Ω through $\mathcal{O}(g^2)$ in pure YM — completing partial $\mathcal{O}(g^2)$ results of Lublinsky–Mullian '17.
- Consistent treatment of **operator-valued, non-commuting** ρ throughout (no classical-source shortcut).
- Perturbation theory reformulated for **non-commuting matrix elements**; the $[\rho, \rho]/\{\rho, \rho\}$ decomposition as a systematic ρ -counting.

Conceptual

- Ω has a **dual role**: it generates the LCWF *and* diagonalises H .
- $\mathcal{O}(g)$ diagonalisation needs **both** the coherent (WW) operator *and* the $3g$ operator C .
- At $\mathcal{O}(g^2)$ only the ρ^2 background energy survives — a non-trivial consistency statement.
- Beyond LO, Ω contains **connected** multi-gluon amplitudes at fixed ρ .

Where it goes next

Single inclusive gluon production at NLO at mid-rapidity (also needs some $\mathcal{O}(g^3)$ coefficients from LM '17) • moments of the gluon **multiplicity operator** and number **fluctuations** at $\mathcal{O}(g^4)$ • double / triple inclusive production and the near-side ridge beyond the coherent state.

Summary

1. **LCPT** is ordinary time-ordered perturbation theory on null planes. Positivity of p^+ makes the vacuum trivial, so the **Fock expansion of a hadron is a physical object**, and longitudinal boosts are kinematical. Cost: $A^+ = 0$ gauge brings **instantaneous $1/\partial^+$ vertices**.
2. **In the CGC**, boosting a hadron by Δy is a **unitary operator Ω** acting on the soft Fock space, with coefficients that are functionals of the **operator-valued** valence charge density ρ . At LO Ω is a coherent state of Weizsäcker–Williams gluons.
3. **This work** fixes **all** coefficients of Ω through $\mathcal{O}(g^2)$ by combining **unitarity $\Omega^\dagger \Omega = 1$** with **matching to LCPT** on $|0\rangle, |g\rangle, |gg\rangle, |ggg\rangle$.
4. **Check & payoff**: $\Omega^\dagger H \Omega = H_0 - \int g^2 \rho \rho / (p^+)^2$ — the Hamiltonian is diagonal up to the classical background energy. The same coefficients are the input for **NLO gluon production, multiplicity fluctuations and the ridge**.

Thank you — questions?

Backup

Backup: LCPT matrix elements used

$$\langle g_i^a(k) | H_g | 0 \rangle = \frac{g k_i \rho^a(-\mathbf{k})}{4\pi^{3/2} |k^+|^{3/2}}$$

$$\langle g_i^c(q) g_j^b(p) | H_g g_i^a(k) \rangle = \delta^{ac} \delta_{ij} \delta^{(3)}(k - q) \frac{g p_j \rho^b(-\mathbf{p})}{4\pi^{3/2} |p^+|^{3/2}} + (p \leftrightarrow q, b \leftrightarrow c)$$

$$\begin{aligned} \langle g_i^c(q) g_j^b(p) | H_{ggg} g_i^a(k) \rangle &= \frac{ig f^{abc} \delta^{(3)}(k - p - q)}{8\pi^{3/2} \sqrt{k^+ p^+ q^+}} \left[(p_i - q_i + \frac{q^+ - p^+}{k^+} k_i) \delta_{jl} \right. \\ &\quad \left. + (k_j + q_j - \frac{k^+ + q^+}{p^+} p_j) \delta_{il} + (\frac{k^+ + p^+}{q^+} q_l - p_l - k_l) \delta_{ij} \right] \end{aligned}$$

$$\langle g_j^c(q) g_i^b(p) | H_{gg\text{-inst}} 0 \rangle = \frac{ig^2 f^{abc} (q^+ - p^+) \delta_{ij} \rho^a(-\mathbf{p} - \mathbf{q})}{2(2\pi)^3 \sqrt{p^+ q^+} (p^+ + q^+)^2}$$

$$\langle g_j^c(q) | H_{gg\text{-inst}} g_i^b(p) \rangle = \frac{ig^2 f^{abc} (p^+ + q^+) \delta_{ij} \rho^a(\mathbf{p} - \mathbf{q})}{2(2\pi)^3 \sqrt{p^+ q^+} (p^+ - q^+)^2}$$

Backup: the kinematic window for $a^\dagger a$ transitions

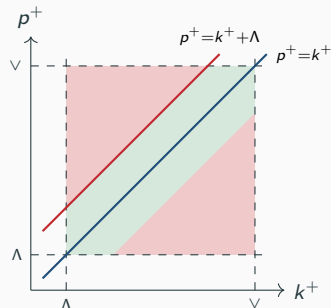
For gluon-number preserving transitions off the valence background, *all three* longitudinal momenta must sit in the soft window:

$$\Lambda < k^+ < V, \quad \Lambda < p^+ < V,$$

$$\Lambda < |p^+ - k^+| < V.$$

This produces the Θ -functions in B_3 :

$$\Theta(p^+ - k^+ - \Lambda) + \Theta(k^+ - \Lambda - p^+).$$



The intersection of the shaded regions is the allowed p^+ range; the two Θ -functions correspond to p^+ above / below the incoming k^+ , i.e. to longitudinal momentum flowing out of / into the valence sector.

Backup: perturbation theory with non-commuting matrix elements

Standard Rayleigh–Schrödinger, but the matrix elements are **operators in the valence Hilbert space** and must be kept in order:

$$|n\rangle = |n^{(0)}\rangle - \sum_{i \neq 0} \frac{|n^{(i)}\rangle \langle n^{(i)}| H_{\text{int}} |n^{(0)}\rangle}{E_i^{(0)} - E_0^{(0)} - E_i^{(1)} - E_i^{(2)}} + \sum_{i,j \neq 0} \frac{|n^{(i)}\rangle \langle n^{(i)}| H_{\text{int}} |n^{(j)}\rangle \langle n^{(j)}| H_{\text{int}} |n^{(0)}\rangle}{(E_i^{(0)} - E_0^{(0)} - E_i^{(1)})(E_j^{(0)} - E_0^{(0)} - E_j^{(1)})}$$

with $E_i^{(1)} \equiv \langle n^{(i)}| H_{\text{int}} |n^{(i)}\rangle$.

Why it matters

For $E_0^{(0)} = 0$ (the soft vacuum),

$$|\psi\rangle_N = |0\rangle - \frac{|n^{(i)}\rangle \langle n^{(i)}| H_{\text{int}} |0\rangle}{E_i^{(0)}} + \frac{|n^{(i)}\rangle \langle n^{(i)}| H_{\text{int}} |n^{(j)}\rangle \langle n^{(j)}| H_{\text{int}} |0\rangle}{E_i^{(0)} E_j^{(0)}}.$$

Writing $\langle i| H |j\rangle \langle j| H |0\rangle$ in the *wrong* order would change the colour structure at $\mathcal{O}(g^2)$ — this is the difference between $[\rho, \rho]$ and $\{\rho, \rho\}$ contributions.

Backup: dictionary for the equal-time-minded

	equal time ($t = 0$)	light front ($x^+ = 0$)
Hamiltonian	P^0	P^-
Dispersion	$E = \sqrt{\mathbf{p}^2 + m^2}$	$p^- = \frac{\mathbf{p}^2 + m^2}{2p^+}$
Vacuum	non-trivial condensate	trivial (perturbatively)
Boost along z	dynamical	kinematical
Gauge	$A^0 = 0$ (Coulomb term)	$A^+ = 0$ (inst. $1/\partial^+$)
Fock expansion	frame dependent	frame independent
Ghosts	yes (covariant gauges)	no
Pathologies	—	$p^+ \rightarrow 0$ zero modes

The trade

Light-front quantisation buys a probabilistic, boost-invariant description of the hadron. It pays for it with a singular $p^+ \rightarrow 0$ region — which in the small- x context is not a bug: the $\ln(\sqrt{s}/\Lambda)$ divergence is the rapidity evolution.